

Razvan Florian Vasile

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EDUCATION

Master's Degree in Computer Science

2022 - 2025

University of Bologna (Unibo), Bologna, Italy – Awarded March 2026

- Completed a Master's degree in Computer Science with a focus on Machine Learning.
- Final grade: 110/110 with honours.

BSc (Hons) Computer Science and Mathematics

2014 - 2018

University of Manchester (UoM), Manchester, UK – Incl. Industrial Experience Year

- Finished a joint honours degree in Computer Science and Mathematics.
- Final grade: First Class Honours.
- Awarded the Manchester Scholarship on entry (A*AA equivalent).

EMPLOYMENT AND EXPERIENCE

Software Engineer

2018 - 2020 / 2021 - 2022

Onit Group / Agronica, Cesena, Italy

Full-stack developer · Python · Databases · IoT

- Developed Raspberry Pi IoT integrations in Python for warehouse equipment control.
- Full-stack development across application services supporting SQL/NoSQL databases.

Industrial Placement Year

2016 - 2017

Diamond Light Source, Didcot, UK

Accelerator controls · Python · MATLAB

- Developed a unified library for accelerator controls, creating a single Python interface to access previously fragmented MATLAB-based accelerator components.
- Enabled programmatic experimentation in Python, allowing precise control of magnet steering positions to optimize beam trajectories for diverse experimental setups.

Peer Assisted Study Sessions (PASS) Leader

2015 - 2016

University of Manchester Teaching Assistant, UK

- Assisted first-year students with programming concepts and coursework problems.
- Organised review sessions and prepared practice materials for exams.

Hornet Representative

2014 - 2016 / 2017 - 2018

University of Manchester IT Services, UK

IT Support · Customer Service · Hardware & Software

- Resolved wired and wireless connectivity issues across university halls of residence.
- Visited residents on site to diagnose and resolve network connectivity problems.

TECHNICAL SKILLS

Languages: Python, C++, MATLAB, JavaScript/TypeScript.

Experienced with: Synthetic Data Generation, Distillation, SFT, RLVR via GSPO, LoRA/PEFT, RAG, Embeddings & Vector Search, Multimodal Training, vLLM, Docker.

Math. Background: Linear Algebra, Matrix Analysis, Convex Optimization, PDEs & Calculus.

Exposure to: PyTorch (DDP), CUDA, cuDNN, cuBLAS, GPU kernels, Profiling.

References and more information available upon request.

PROJECTS

Master's Thesis: Distilling Neurosymbolic Reasoning for Linear Algebra in Small Language Models

2025 – 2026

Tool-using agents · Synthetic data · Distillation · SFT · RLVR

[GitHub](#) · [Report](#) · [Poster](#) · [Demo](#) · [HF Space](#)

- Built an end-to-end neurosymbolic pipeline for training the Qwen2.5-3B pre-trained base model to solve linear algebra problems via sequential tool calls.
- Implemented synthetic data generation, knowledge distillation, SFT, and RLVR/GSPO via LoRA, relying on deterministic reward functions.
- Achieved 90.26% end-to-end accuracy on the test set using \$75 of cloud compute.

Bachelor's Thesis: Solving Sudoku using Propositional Logic

2017 – 2018

C++ · Propositional Satisfiability · SAT Solver

[GitHub](#) · [Report](#) · [Presentation](#)

- Implemented a CDCL SAT solver in C++ including backjumping, conflict-driven clause learning, two SAT encodings and two decision heuristics.
- Built an interactive Sudoku solver to visualize the resolution process via implications graphs, illustrating backpropagation and decision semantics.
- Evaluated solver performance on synthetic SAT instances via different decision heuristics (i.e., VSIDS/LIS) and profiled encodings (i.e., minimal, extended) via flame graphs.

Distributed and Parallel Techniques for Deep Neural Networks

2025

Survey · PyTorch DDP · CUDA · Docker

[GitHub](#) · [Report](#) · [Poster](#)

- Conducted a literature review of distributed training and GPU libraries, comparing parallelization strategies and implementation trade-offs.
- Built GPU training experiments using CUDA (cuDNN and cuBLAS), covering forward and backward passes, convolution primitives, manual kernels and linear layers.
- Produced reproducible distributed-training examples with PyTorch DDP and Docker to simulate multi-GPU and multi-node workflows on a single GPU.

Visual QA: Using Generative Models on Classification Tasks

2024

Multimodality · BLIP-2 · LoRA · UMAP

[GitHub](#) · [Report](#) · [Poster](#)

- Adapted the BLIP-2 multimodal architecture for discriminative tasks by modifying the existing Q-Former architecture.
- Fine-tuned the model with LoRA, achieving 91% accuracy on Easy-VQA and 78% on DAQUAR datasets, outperforming generative baselines.
- Used UMAP projections of the learned embeddings to verify the generation of semantically appropriate per-class clusters.

Git Inspector: Querying GitHub Repositories with Local LLMs

2025

Semantic retrieval · REST API · Integration tests

[GitHub](#) · [Report](#) · [Poster](#) · [Website](#) · [Demo](#)

- Built a developer tool based on a RAG system for querying GitHub repositories using semantic code retrieval with Qdrant and local open-source LLMs via Ollama.
- Implemented repository cloning, indexing, semantic chunking and a REST API with a Scala.js front-end interface for code and documentation queries.
- Validated the system through acceptance tests and a usability study that achieved an average SUS score of 85%.